

The Program in Action

Figure 27-1 show what the output will look like when you run *fishtank.py*.

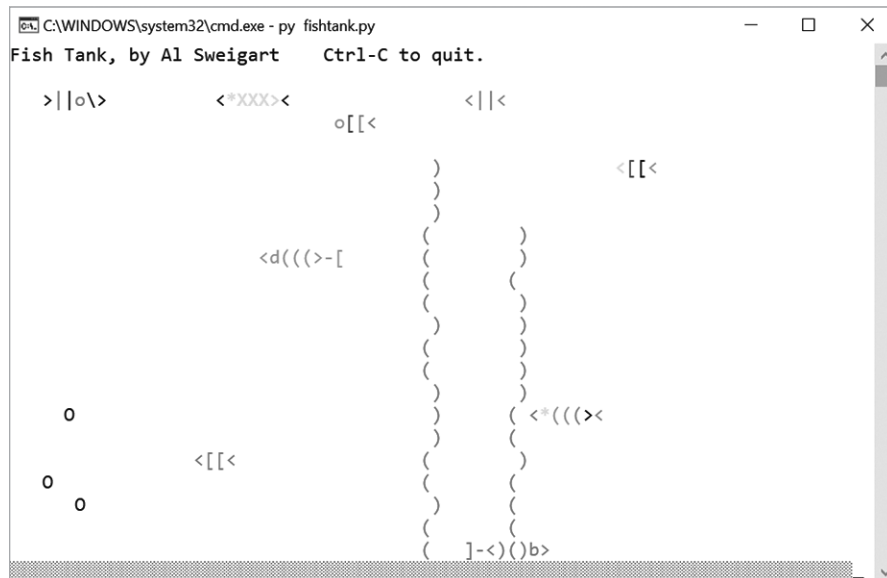


Figure 27-1: The output of the fish tank program, with several fish, kelp plants, and bubbles

How It Works

Modern graphical programs often generate animations by erasing their entire window and redrawing it 30 or 60 times a second. This gives them a *frame rate* of 30 or 60 frames per second (FPS). The higher the FPS, the more fluid the animated movement appears.

Drawing to terminal windows is much slower. If we erased the entire terminal window to redraw its contents with the `text` module, we typically would only get about 3 or 4 FPS. This would cause a noticeable flicker in the window.

We can speed this up by only drawing characters to the parts of the terminal window that have changed. Most of the fish tank program's output is empty space, so to make the elements move, the `clearAquarium()` only has to draw ' ' space characters to the places where the fish, kelp, and bubbles currently are. This increases our frame rate, reduces flickering, and makes for a much more pleasant fish tank animation.

1. `"""Fish Tank, by Al Sweigart al@inventwithpython.com`
2. `A peaceful animation of a fish tank. Press Ctrl-C to stop.`
3. `Similar to ASCIIQuarium or @EmojiAquarium, but mine is based on an`
4. `older ASCII fish tank program for DOS.`
5. `https://robobunny.com/projects/asciiquarium/html/`
6. `https://twitter.com/EmojiAquarium`